

Skills Summary

Around or Across? A Field Guide

Use this reference while completing your worksheet or when studying.

Circumference and Area of a Circle

Formulas: circumference $C = \pi d = 2\pi r$ and area $A = \pi r^2$.

The **diameter is twice the radius**: $d = 2r$. Before using any formula, ask: *was I given r or d ?* The area formula only ever takes r .

Example: A circle has radius 5 cm. Find its circumference.

Step 1. We were given the *radius* and want the distance around — use $C = 2\pi r$.

Step 2. $C = 2\pi \cdot 5 = 31.415\dots$

$\Rightarrow$ **31.42** cm

Try it: A circle has *diameter* 18 cm. Find its circumference.

check: 56.55 cm

Perimeter or Area? Ask What the Question Buys

Rule for rectangles: perimeter $P = 2(\ell + w)$ measures the distance *around* — fences, frames, trim. Area $A = \ell \cdot w$ measures the surface *covered* — paint, carpet, sod. Perimeter answers come in m or ft; area answers come in m^2 or ft^2 .

Example: A yard is 8 m by 3 m. How much fencing surrounds it, and how much sod covers it?

Step 1. Fencing goes *around* (perimeter); sod *covers* (area) — two different formulas on the same numbers.

Step 2. $P = 2(8 + 3) = 22$ and $A = 8 \cdot 3 = 24$.

Fencing: **22** m Sod: **24** m^2

Try it: A frame surrounds a 6 in by 4 in photo. How many inches of frame?

check: 20 in

Catching Radius-vs-Diameter Mistakes

Checklist: (1) If the area came out about *four times* too big, the diameter probably went in where the radius belonged. (2) If a circumference came out exactly double or half what a classmate got, one of you mixed up $C = \pi d$ and $C = 2\pi r$. (3) An "area" measured in plain meters or feet is a perimeter in disguise.

Example: Kim says a circle with diameter 20 m has area $\pi \cdot 20^2 = 1256.64$ square meters. Fix her work.

Step 1. The area formula takes the *radius*, and the radius is half of 20, which is 10.

Step 2. $A = \pi \cdot 10^2 = 314.159\dots$ — one quarter of Kim's value.

$\Rightarrow$ **314.16** m^2

Try it: Find the area of a circle with *diameter* 8 in.

check: 50.27 in²